

Looped Propeller Design Aims to Reduce Noise Near Whales

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A Detroit-based company called Sharrow Marine is partnering with marine conservation organization Ocean Alliance to test quieter boat propellers on research **expeditions** near whales. The work is still at an early stage, but it could help explore a way to reduce underwater noise from boats.

Noise pollution is not limited to crowded streets. Large shipping boats, deep-sea mining equipment and offshore oil rigs have changed the ocean's sound environment. Whales rely on calls to coordinate with one another, find food and locate mates. When the water is quiet, they can hear calls from hundreds of miles away. However, commercial shipping produces noise that overlaps with the **low-frequency** sounds whales use to communicate.

Michelle Fournet, an acoustic ecologist and ocean noise pollution expert at the University of New Hampshire, described this interference to Popular Science. She compared persistent ocean noise to someone repeatedly using a jackhammer outside a window. Research described in the source also linked louder noise around orcas with a lower chance of catching fish they chased.

Much of the disruptive noise from boat propellers comes from **cavitation**. When a screw-shaped propeller spins quickly, tiny bubbles form behind it. As these bubbles pop, they release energy and create hissing or crackling sounds. A single propeller can produce and pop millions of bubbles per second.

Sharrow's design replaces blades with free-standing tips with looped blades that twist like a ribbon. The company says the design produces fewer tip **vortices**, or swirling flows at the blade tips, reducing **cavitation**. It claims independent testing showed underwater noise reductions of up to seven **decibels** in the frequency range whales use to communicate. **Decibels** measure sound levels. These are company-reported findings; the source does not provide the full testing methods or results.

Greg Sharrow, the company's founder and CEO, says the amount of noise reduction varies with frequency, application and operating conditions. Measurements above water also cannot establish how much quieter a boat sounds to whales underwater. That distinction matters when assessing possible benefits for marine animals.

The idea originally grew from Sharrow's work recording orchestras. He wanted to film a violinist using a camera drone inside an orchestra, but its buzzing interfered with the music. Thinking about a propeller without separate blade tips led him to consider boats too. Early **prototypes** were tested in a fish tank. He says he made more than 640 **prototypes** before partnering with the University of Michigan, which led to testing on a small boat.

Ocean Alliance CEO Iain Kerr supports approaches that benefit the environment, animals and boat users. Greg Sharrow believes quieter vessel noise could help marine animals hear one another with less interference. The research partnership is exploring the design near whales; benefits for their communication have not yet been established in the supplied

report.

Word Treasure

expeditions -- Trips made for a purpose, such as research.

low-frequency -- Describes sounds with slow vibrations and a low pitch.

cavitation -- The forming and popping of tiny bubbles near a propeller.

vortices -- Swirling flows of water or air.

decibels -- Units used to measure sound levels.

prototypes -- Early versions of a design made for testing.

Background you need

Whales rely on calls to coordinate, find food and locate mates.

Cavitation bubbles release energy and produce sound when they pop.

The company says noise reductions depend on frequency, application and operating conditions.

Break it down

Introduces the partnership to test quieter propellers near whales.

Explains underwater noise and its interference with whale calls.

Describes cavitation and the proposed looped-blade solution.

Attributes test claims to the company and explains measurement limits.

Traces the design from orchestra filming to boat prototypes.

Ends with attributed hopes and the limits of the reported evidence.

Why it matters

The design offers a possible way to reduce boat noise that overlaps with whale calls.

Underwater testing matters because quieter sound above water alone cannot establish benefits for whales.

Test what you remember

1. Which organization is partnering with Sharrow Marine on research trips?
 - (A) University of New Hampshire
 - (B) Ford Motor Company
 - (C) University of Michigan
 - (D) Ocean Alliance
2. Why can commercial shipping noise interfere with whale communication?
 - (A) It increases the depth whales swim at.
 - (B) It changes the routes whales travel.
 - (C) It overlaps with the sounds whales use.
 - (D) It removes the food whales search for.
3. According to the company, how do looped blades reduce cavitation?
 - (A) They produce fewer tip vortices.
 - (B) They move boats without water.
 - (C) They stop all propeller movement.
 - (D) They collect bubbles in a tank.
4. What underwater noise reduction does the company report in whales' communication range?
 - (A) Up to fifteen decibels
 - (B) Up to seven decibels
 - (C) Up to twenty-four decibels
 - (D) Up to twenty decibels
5. Why does the article distinguish measurements above water from those underwater?
 - (A) Above-water results cannot establish what whales hear underwater.
 - (B) Above-water results can replace research expeditions near whales.
 - (C) Above-water results can explain every change in whale calls.
 - (D) Above-water results cannot measure any noise from boat propellers.
6. Where were Greg Sharrow's early propeller prototypes tested?
 - (A) Inside an orchestra hall
 - (B) Near a shipping port
 - (C) Near an oil rig
 - (D) Inside a fish tank

Answer key (try the quiz first!): 1: D 2: C 3: A 4: B 5: A 6: D

Questions to think about

1. **Michelle Fournet:** She describes persistent ocean noise as disruptive, comparing it to repeated jackhammer noise outside a window.
2. **Greg Sharrow:** He believes reducing vessel noise where it overlaps with animal communication could help marine animals hear one another.
3. **Iain Kerr:** He supports approaches that benefit the environment, animals and boat users.

MY THOUGHTS (at least 20 words):